

Olist E-Commerce Knowledge Guide

Reference document for the Olist Brazilian E-Commerce dataset (2016–2018)

This guide describes how the Olist marketplace works, what the dataset contains, and how customers, sellers, products, orders, payments, and reviews relate to one another. It is meant to answer descriptive, "how does this work" style questions. Exact numbers, totals, and statistics (revenue, order counts, averages, etc.) should be answered from the structured dataset/database directly, not from this document.

1. Platform Overview

Olist is a Brazilian marketplace technology company that connects small and medium-sized merchants ("sellers") to major Brazilian online marketplaces, allowing them to sell products under a single contract instead of registering separately with each channel. Sellers list their products through Olist, and once a customer places an order, the seller is notified and ships the product directly to the customer using Olist's logistics partners.

This dataset reflects **real, anonymized commercial orders placed between 2016 and 2018**, covering the full journey of an order — from purchase, to payment, to shipping, to delivery, and finally to the customer's post-purchase satisfaction survey.

Core entities in the dataset:

- **Customers** — buyers identified by a customer ID and location (city, state, zip code prefix).
- **Sellers** — merchants identified by a seller ID and location, responsible for fulfilling orders.
- **Products** — items for sale, each with a category, physical dimensions, weight, and listing details (name length, description length, number of photos).
- **Orders** — the central record of a transaction, tracked through several status stages with timestamps.
- **Order items** — individual line items within an order (an order can contain multiple products, potentially from different sellers).
- **Payments** — one or more payment records per order, capturing method, installments, and value.
- **Reviews** — post-purchase satisfaction ratings and comments submitted by customers.
- **Geolocation** — latitude/longitude data linked to Brazilian zip code prefixes, used for mapping customer and seller locations.

An order can be understood as a pipeline: **Order Placed** → **Payment Approved** → **Processing** → **Shipped** → **Delivered** → **Reviewed**. Each stage leaves a trace in the data (timestamps, status flags, and eventually a review score), which is how order experience and delays can be analyzed.

2. Product Catalog and Categories

The catalog spans a wide range of everyday consumer goods. Product categories are recorded in Portuguese in the raw data and mapped to English through a translation table. Common category groups include:

- **Home & furniture** — bed_bath_table, furniture_decor, housewares, home_construction
- **Office** — office_furniture, stationery
- **Electronics & computers** — computers_accessories, electronics, telephony, computers, audio
- **Health & beauty** — health_beauty, perfumery
- **Sports & leisure** — sports_leisure
- **Toys & baby** — toys, baby, cool_stuff
- **Fashion** — fashion_bags_accessories, fashion_shoes, fashion_underwear_beach, fashion_male_clothing
- **Auto & tools** — auto, garden_tools, construction_tools_construction
- **Books, music & stationery** — books_general_interest, books_technical, music, cds_dvds_musicals
- **Food & drink** — food, drinks, food_drink

There are over 70 distinct product categories in total, reflecting a broad department-store-style catalog rather than a narrow specialty shop. Each product record also includes physical attributes — weight in grams, and length/height/width in centimeters — which are used to calculate shipping (freight) cost.

Listing quality fields tracked per product include the length of the product name, the length of the product description, and the number of photos included in the listing. These fields reflect how completely a seller has documented their product, which in practice correlates with buyer confidence and conversion.

3. Order Lifecycle and Status

Every order moves through a defined set of statuses as it progresses:

Status	Meaning
created	Order has been placed but payment has not yet been confirmed.
approved	Payment has been approved/confirmed.
processing	Seller has been notified and is preparing the order.
invoiced	An invoice has been issued for the order.
shipped	The order has been handed off to a logistics carrier.
delivered	The order has reached the customer.
unavailable	The order could not be fulfilled (e.g., stock issue).
canceled	The order was cancelled at some point in the process.

Along with status, each order carries a set of **timestamps**: purchase time, approval time, the date it was handed to the carrier, the estimated delivery date, and the actual delivery date. The gap between the estimated and actual delivery date is a key measure of delivery performance and customer satisfaction, and it is one of the strongest drivers of low review scores when a delivery arrives later than promised.

Delays typically stem from: slow payment approval, seller processing time, carrier transit time, incorrect or hard-to-reach customer addresses, and regional logistics conditions across Brazil's states.

4. Payments

Each order has one or more associated payment records. Recorded payment types include:

- **Credit card** (**credit_card**) — the most commonly used method, often paid in installments.
- **Boleto** (**boleto**) — a Brazilian bank-slip payment method where the customer pays at a bank, lottery outlet, or online banking portal; confirmation can take longer than card payments since it depends on the customer completing the payment step.
- **Debit card** (**debit_card**) — processed similarly to credit card but drawn directly from available funds.
- **Voucher** (**voucher**) — store credit or voucher-based payment, sometimes combined with another method to cover part of an order.

Payments also record the **number of installments**, which is common for credit card purchases in the Brazilian market, and the **payment value**, which may be split across multiple payment records if a customer used more than one method for a single order.

An order generally cannot move into seller processing until its payment has been approved. If payment fails or is never completed, the order is typically cancelled.

5. Customer Reviews and Satisfaction

After a delivery is completed (or the estimated delivery date has passed), the customer receives a satisfaction survey by email. This produces a **review** linked to the order, consisting of:

- **Review score** — a rating from 1 (very dissatisfied) to 5 (very satisfied).
- **Review comment title and message** — optional free-text feedback from the customer.
- **Review timestamps** — when the survey was sent and when the customer responded.

Review scores are the primary signal of customer experience in the dataset. Low scores are strongly associated with late deliveries, damaged or incorrect items, and poor communication, while high scores are associated with on-time or early delivery and products that matched their listing.

6. Returns, Refunds, and Order Cancellations

The dataset does not contain a dedicated "returns" or "refund request" table. Refund-and-return activity is instead reflected indirectly through:

- **Order status** — an order marked **canceled** or **unavailable** indicates the transaction did not complete as originally planned.
- **Review scores and comments** — customers frequently describe issues such as receiving the wrong item, a damaged product, or a very late delivery in their review text, which is the closest signal to a complaint or dissatisfaction event in the data.

In general marketplace practice (and as the operating assumption for this guide), a customer seeking a refund would report the issue — wrong item, damaged item, or non-delivery — through customer support, and a resolution (refund or replacement) would be issued once the claim is reviewed and, where applicable, once a returned item is received back from the customer. Because this dataset does not track that support workflow directly, questions like "can I get a refund" should be answered descriptively (explaining how such requests are typically handled) rather than by citing a specific refund total or count, which the data does not capture.

7. Sellers and Marketplace Operations

Sellers are independent merchants, each tied to a location (city, state, zip code prefix). A single order can include items from multiple sellers, since Olist aggregates listings from many independent merchants into one customer-facing checkout experience.

Seller-side factors that show up in the data and affect the customer experience include:

- How quickly an order moves from "processing" to "shipped" (processing speed).
- Whether the shipped item matches the listing (reflected in review comments).
- Freight value relative to product price, which depends on the seller's location relative to the customer and the product's weight/dimensions.

Because delivery experience is a shared outcome of seller handling, carrier performance, and geography, seller location and shipping distance are useful factors to consider together when interpreting delays or low review scores.

8. Business Knowledge FAQ

Q: What kinds of products does Olist sell? A: A broad department-store-style catalog spanning home and furniture, electronics, health and beauty, sports and leisure, toys, fashion, books, auto, and more — over 70 product categories in total.

Q: Can I get a refund? A: Refunds are generally handled by reporting an issue (wrong item, damaged item, or non-delivery) to support, after which the order is reviewed and a refund or replacement is issued. The dataset reflects this indirectly through cancelled orders and negative review comments rather than a dedicated refund log.

Q: What does it mean when an order status is "canceled"? A: The order was not completed — it may have failed at payment, been cancelled by the customer, or become unavailable before shipment.

Q: What payment methods can customers use? A: Credit card, debit card, boleto (a Brazilian bank-slip method), and vouchers; credit card payments can also be split into installments.

Q: Why do some payments take longer to confirm? A: Boleto payments require the customer to complete an extra step (e.g., paying at a bank), so confirmation is slower than with card payments.

Q: How is customer satisfaction measured? A: Through a post-delivery review survey, where the customer gives a score from 1 to 5 and can leave a comment.

Q: What usually causes a low review score? A: Late delivery relative to the estimated delivery date, damaged or incorrect products, and poor communication are the most common drivers.

Q: What information is tracked about each product? A: Category, weight, dimensions (length, height, width), and listing quality details like name length, description length, and photo count.

Q: Can one order include products from multiple sellers? A: Yes — since Olist aggregates independent sellers into one checkout, an order can contain items fulfilled by different sellers.

Q: What determines shipping (freight) cost? A: Primarily the product's weight and dimensions, combined with the distance between the seller's and customer's locations.

Q: What is the difference between the estimated and actual delivery date? A: The estimated delivery date is the date promised to the customer at checkout; the actual delivery date is when the order truly arrived. The gap between them is a key measure of delivery performance.

Q: What happens after a customer places an order? A: Payment is submitted for approval; once approved, the order moves to processing, then shipping, then delivery, followed by the satisfaction survey.

Q: Does Olist sell products directly? A: No — Olist is a marketplace connecting independent sellers to customers; sellers own the inventory and fulfill orders themselves.

Q: What does the "unavailable" order status mean? A: The order could not be fulfilled, typically due to a stock or availability issue on the seller's side.

Q: What geographic information is available? A: Customer and seller locations are recorded at the city/state/zip-prefix level, with a separate geolocation table linking zip prefixes to latitude/longitude coordinates.

9. Glossary

Customer — The buyer who places an order, identified by location and a customer ID.

Seller — An independent merchant fulfilling orders through the Olist marketplace. **Order** — The central transaction record, tracked through a defined sequence of statuses. **Order item**

— A single product line within an order; one order may contain several. **Payment** — A

record of how and how much was paid for an order, including method and installments. **Review** — A post-delivery satisfaction rating (1–5) and optional comment from the customer. **Order status** — The current stage of an order (created, approved, processing, invoiced, shipped, delivered, unavailable, canceled). **Estimated delivery date** — The delivery date promised to the customer at checkout. **Freight value** — The shipping cost associated with an order item. **Boleto** — A Brazilian bank-slip payment method requiring an extra confirmation step. **Product category** — The classification of a product (e.g., electronics, toys, furniture) used for browsing and search. **Geolocation** — Latitude/longitude data linked to Brazilian zip code prefixes.